

Assignment 3

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Q1:

c) Train

n	Model 1	Model 2
16	0.9125	0.9125
32	0.928125	0.99375
64	0.8765625	0.965625
128	0.8640625	0.95859375

Test

n	Model 1	Model 2
16	0.7108	0.8804
32	0.7948	0.8933
64	0.827	0.906
128	0.8369	0.9114

Q2:

e) Train

Dim k	Model 1	Model 2
1	0.84984375	0.6371875
2	0.85453125	0.94195312

Test

Dim k	Model 1	Model 2
1	0.84261	0.62448
2	0.83791	0.91636

Q3:

c)

k	2	3	4
obj	2.187413817121596	1.32537303194761	0.7731572415061299
k	5	6	7
obj	0.643662759610057	0.5344011976444857	0.4313001442935195
k	8	9	
obj	0.38525582966742994	0.3453978158601691	

As we can see after cluster 4 changes becomes much less significant. Thus 4 clusters would be appropriate

cont...

d)

Best objective value: 0.4183586786341489

